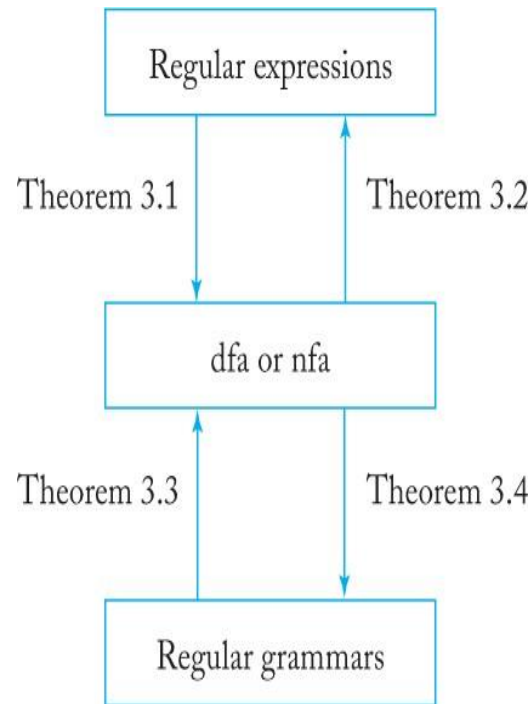


Automata

languages lexers, parsers, interpreters



Regular languages and regular grammars



Chapter 3 consists largely of definitions and proofs of the above scheme

Definition (chapter 2)

A language L is called **regular** if and only if there exists some deterministic finite acceptor M such that

$$L = L(M).$$

Regular expression

- A way of describing regular languages
- A combination of strings of symbols from some alphabet Σ , parentheses, and the operators $+$, $\cdot$, and $*$.
- Sample
 - $a + b + c \rightarrow$ Language $\{a,b,c\}$
 - $(a + (b \cdot c))^* \rightarrow$ language $\{\lambda, a, bc, aa, abc, bca, bc bc, aaa, aabc, \dots\}$.

Definition regular expression

Let Σ be a given alphabet. Then

1. $\emptyset$, λ , and $a \in \Sigma$ are all regular expressions. These are called primitive regular expressions.
2. If r_1 and r_2 are regular expressions, so are $r_1 + r_2$, $r_1 \cdot r_2$, r_1^* , and (r_1) .
3. A string is a regular expression if and only if it can be derived from the primitive regular expressions by a finite number of applications of the rules in (2).

Right- and Left-Linear Grammars

A grammar $G = (V, T, S, P)$ is said to be **right-linear** if all productions are of the form

$$A \rightarrow xB \text{ or } A \rightarrow x,$$

where $A, B \in V$, and $x \in T^*$.

A grammar is said to be **left-linear** if all productions are of the form

$$A \rightarrow Bx \text{ or } A \rightarrow x$$

A **regular grammar** is one that is either right-linear or left-linear

Sample

The grammar $G1 = (\{S\}, \{a, b\}, S, P1)$, with $P1$ given as $S \rightarrow abS|a$ is right-linear.

The grammar $G2 = (\{S, S_1, S_2\}, \{a, b\}, S, P2)$, with $P2$ given as $S \rightarrow S_1ab$, $S_1 \rightarrow S_1ab|S_2$, $S_2 \rightarrow a$, is left-linear. Both $G1$ and $G2$ are regular grammars.

The sequence $S \Rightarrow abS \Rightarrow ababS \Rightarrow ababa$ is a derivation with $G1$. From this single instance it is easy to conjecture that $L(G1)$ is the language denoted by the regular expression $r = (ab)^* a$.

In a similar way, we can see that $L(G2)$ is the regular language $L(aab(ab)^*)$.